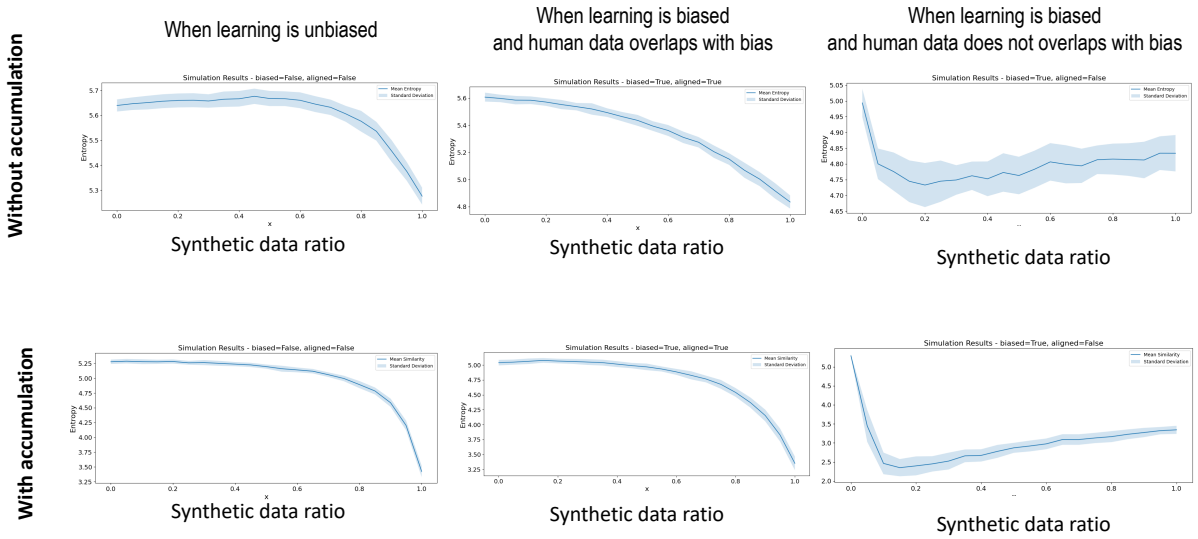
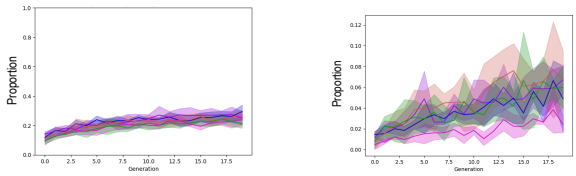


Figure 12: **Simulation results of the toy model.** Under specific conditions, a non-linear u-shaped relationship between diversity loss and synthetic-data ratio emerges.

$\mathbf{r^2}$	All	Webis		100M Tweets		Reddit Submissions		Wikipedia	
Synthetic data ratio		1/8	1/4	1/8	1/4	1/8	1/4	1/8	1/4
Semantic Diversity	0.293	0.244	0.541	0.043	0.188	0.109	0.502	0.147	0.105
Quality	0.737	0.482	0.353	0.193	0.324	0.143	0.564	0.249	0.276

Table 3: R^2 results corresponding to Table 1.



(a) Proportion of politically neutral tweets

(b) Proportion of non-political tweets

Figure 13: (a) Proportion of politically neutral tweets increases, implying that models tend to avoid strong political statements. (b) Proportion of non-political tweets marginally increases, implying that the models stay on the topic of politics.

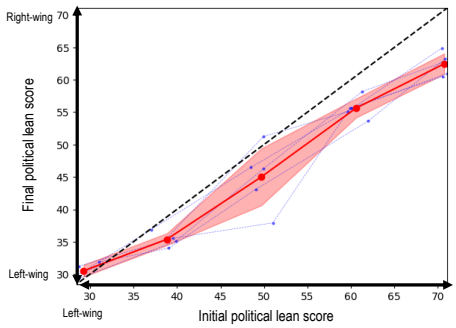


Figure 14: Average political lean at the last generation as a function of political lean in the true distribution after excluding political tweets

Domain	R^2 (Diversity)	R^2 (Quality)
Webis	0.340	0.591
100M Tweets	0.532	0.611
Wikipedia	0.241	0.363

Table 4: R^2 results corresponding to Figure 4.

models may display different biases with respect to political lean, we also ran an experiment with homogeneous chains, where the same base model is used in each generation. To clarify, in each generation the training still starts from a new instance of a pretrained model, e.g. over 20 generations we will initialize 20 separate LLama-3.2-1B instances. Figure 15 compares homogenous chains corresponding to the four considered models. We observe little variation between the chains, although Falcon3-1B seem to display slightly more pronounced left-wing bias, while this bias is weaker for Llama-3.2-1B.

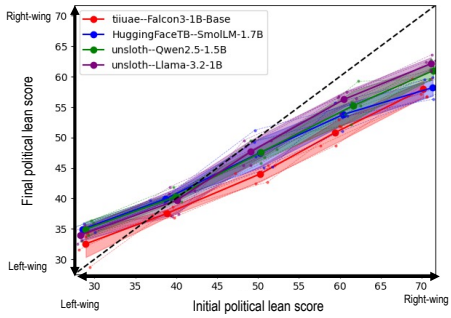


Figure 15: Average political lean at the last generation as a function of political lean in the human distribution with homogeneous transmission chains. We observe slight differences between chain: for instance, Falcon3-1B chain appears to have a stronger left-wing bias than others, while it is weaker for Llama-3.2-1B